

Owned Single-Wire Transactions

Prove a bounded policy before touching a bus

E0 SYNTHETIC REPLAY — NO PIN, PULL-UP, OR SENSOR

Lesson 064 — component
Prerequisites status, fixed storage, modular time
Input copied requests and line receipts

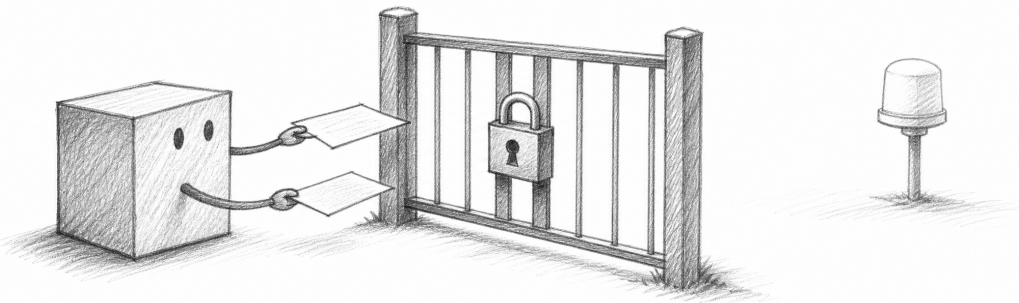
Time 180–240 minutes
Board compile-only Mega replay
Observation named intent/snapshot cells

Responsibility. `OneWireTransactionPolicy` expands one reviewed typed operation into a bounded sequence of copied line intents, validates one correlated copied receipt at a time, retains an attributable accepted prefix, and requests release during completion or rollback. It cannot drive, sample, power, or release a physical line.

1 Separate protocol policy from electrical execution

The DS18B20 data sheet describes an open-drain 1-Wire interface, reset and presence behavior, read and write slots, ROM commands, supply modes, and device timing limits.¹ Those facts explain what a future qualified adapter must implement. They do not prove that an unidentified kit part, a Mega pin, or a particular pull-up meets the requirements.

Lesson 064 therefore stops at a copied evidence boundary. The policy emits semantic values such as `DriveLow`, `Release`, and `Sample`. A synthetic producer returns a copied receipt saying what it applied or observed. E0 proves only that the policy processes that evidence deterministically.



Alternative text: a graphite policy box hands two blank copied cards toward a locked gate; a faint generic sensor silhouette remains isolated beyond the gate, and no physical connection crosses it.

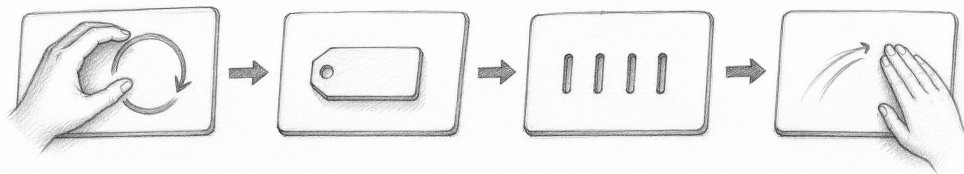
E0 can establish	E0 cannot establish
typed phase order and receipt correlation	pin ownership, voltage, or open-drain behavior
configured timing-window acceptance	physical pulse width, rise time, or interrupt margin
bounded rollback and release intent	a physical line reached idle
byte-stable replay from copied evidence	genuine DS18B20 identity or thermal behavior

There is no formal electrical schematic or wiring exercise in this lesson. An authoritative schematic requires an exact specimen, package and pinout, ordinary pull-up and supply choice, endpoint ownership, measured rails and waveforms, and separate resource and safe-state evidence.

2 Use a closed operation vocabulary

The public request chooses one reviewed operation. Reset/presence, one bounded ROM-search pass, and single-drop ROM read cover identification mechanics. Addressed operations cover power-supply read, conversion start, one conversion status read, and scratchpad read. Their public enum spellings are `ResetPresence`, `SearchRomPass`, `ReadRomSingleDrop`, the four `MatchRom...` operations. No request accepts arbitrary command bytes, captured pulse trains, or an unknown replay script.

¹Analog Devices, *DS18B20 Programmable Resolution 1-Wire Digital Thermometer*, Rev. 6.



Alternative text: four graphite index cards progress through a reset gesture, an identity tag, bounded slot marks, and an open-hand release gesture; the surrounding prose supplies the operation names.

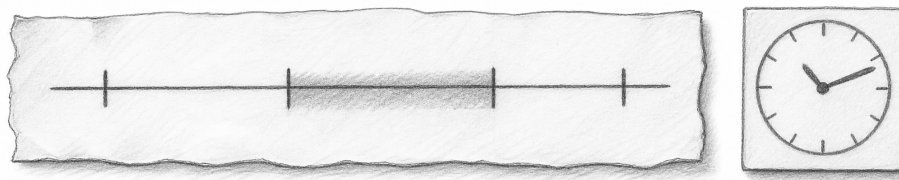
Addressed operations reject only the all-zero ROM. Every other complete 64-bit value is transported byte-for-byte, including a wrong-family or bad-CRC value. Lesson 064 does not call it valid. Lesson 065 exclusively owns family-0x28 and Dallas CRC qualification.

`ReadRomSingleDrop` is admitted only when configuration explicitly names a single-drop fixture. Multidrop work uses exactly one bounded `SearchRomPass`. The manufacturer search algorithm explains why each device's 64-bit ROM participates in a discrepancy tree,² but its enumeration order is evidence order, never physical position.

Why should a wrong-family nonzero ROM be transported rather than "fixed" by Lesson 064? _____.

3 Make microsecond time explicit

The repository's millisecond time type cannot represent single-wire slot windows. Lesson 064 uses a dedicated unsigned 32-bit `MicrosecondTimePoint` and `MicrosecondDuration`. Configuration names independent lower and upper limits for reset low, reset release, presence start, presence low, write-zero low, write-one low, read initiation, read sampling, complete slot, recovery, and the whole transaction deadline.



Alternative text: a graphite timeline on torn paper has four ticks and a shaded admissible window between the middle ticks; a separate clock card represents caller-supplied microsecond time.

Tests inspect immediately below, exactly at, and immediately above every lower and upper boundary. All configured durations stay strictly below the modular half range. Backward time and an exact half-range difference are ambiguous and reject atomically; an ordinary unsigned rollover with a valid forward distance is accepted.

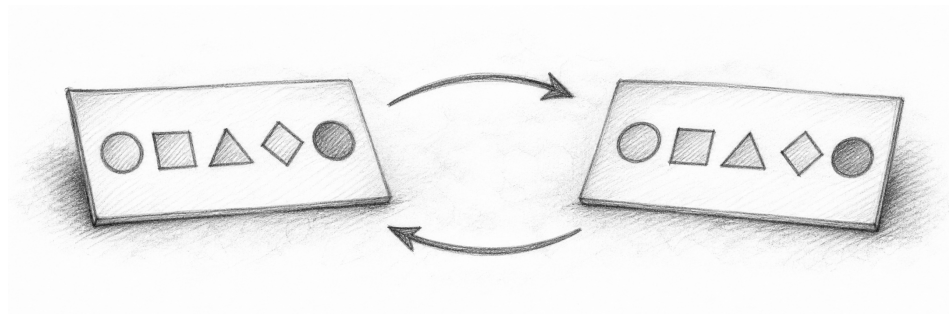
Analog Devices' software guidance emphasizes that recovery and sample timing depend on the device and network conditions.³ The configured E0 windows are copied-evidence acceptance rules, not a claim that any physical waveform met them.

4 Correlate every intent and receipt

An operation request carries request sequence, typed operation, optional addressed ROM or fixed search state, supplied start time, supply mode, and producer status. Every emitted intent binds owner, lifecycle and configuration revisions, request and transaction generations, phase sequence, slot index, operation, line intent, legal action window, and addressed ROM.

²Analog Devices, *1-Wire Search Algorithm*, Application Note 187.

³Analog Devices, *1-Wire Extended Network Standard*, discussing Application Note 126 and network-dependent timing.



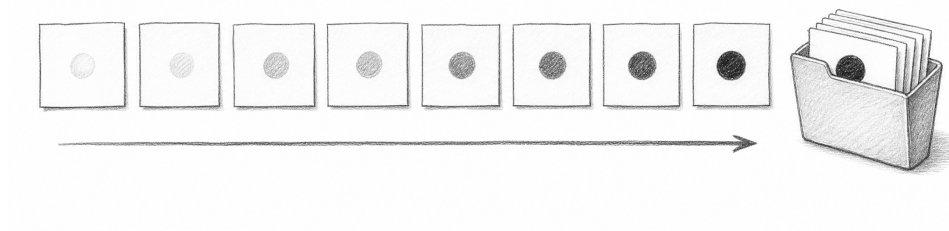
Alternative text: two graphite cards face each other with the same ordered set of geometric seals and opposing arrows, representing exact field correlation and one return receipt.

A matching receipt advances at most one phase or slot. A byte-identical duplicate is idempotent. Changed duplicates, crossed transactions, a foreign source, wrong generation, wrong operation, wrong phase or slot, early/late observation, sequence regression, and exhaustion reject without changing the accepted request, prefix, deadline, or cleanup state.

1. If every field matches except the slot index, should the policy use the sampled bit? _____.
2. If a byte-identical receipt is presented twice, how many phase transitions occur in total? _____.
3. A receipt has valid correlation but producer failure. Does structure reject, or does rollback begin?
_____.

5 Serialize least-significant bits first

The DS18B20 protocol sends data and commands least-significant bit first. Lesson 064 freezes the request at **begin()**, expands the allowlisted operation into bounded private slot mechanics, and assembles copied read bits back into bytes in the same order.



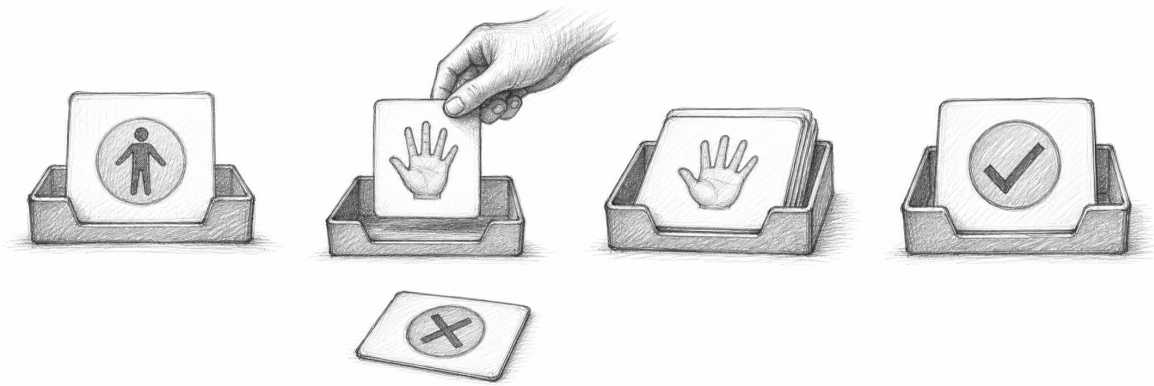
Alternative text: eight graphite cards with progressively darkened dots run left to right into a folder, representing least-significant first slot order and an attributable accumulated prefix.

Partial read bytes are prefix evidence, not a completed device response. A failure at the first, middle, or final slot preserves exactly the admitted prefix and moves to cleanup. Neither an incomplete scratchpad nor a copied search branch becomes a temperature.

For copied bits 1,0,1,1,0,0,0,1 in slot order, write the assembled byte in binary: _____. Which bit becomes the most significant bit? _____.

6 Treat release as an owed transaction

The policy never emits “drive high.” Its canonical inactive and cleanup line intent is **Release**. Initialization, reset, cancellation, timeout, producer failure, sequence exhaustion, and shutdown may all leave one correlated release receipt owed.



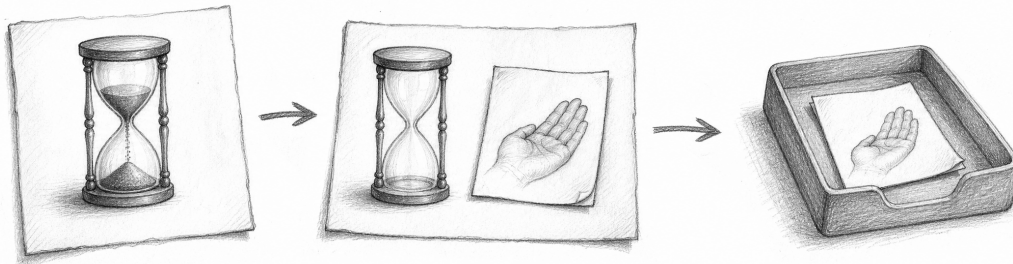
Alternative text: four separated graphite trays show active work, placement of an open-hand release card, a pending stack, and a confirmed card; a failed card lies separately below and does not overlap the path.

Evidence	Permitted statement
policy emitted Release	software requested line release
matching copied release receipt	producer reported applying the request
future E1 waveform observation	qualified physical line reached its idle window

`confirmCleanup()` is the sole closing seam. Reset becomes ready only after its matching copied release receipt. Shutdown remains initialized in a closing state so it can accept that receipt, then becomes inert. Without the receipt, the snapshot remains **ReleaseUnconfirmed**; E0 never says the physical bus is idle.

7 Advance deadlines without polling

Ordinary phase transitions require one receipt through `update()`. The receipt-free `advance(now,intent)` path does only one job: inspect supplied time. Before the deadline it leaves the owed phase unchanged. At the deadline crossing it emits at most one correlated release intent. It never polls, retries, catches up, invents phase evidence, or synthesizes a cleanup receipt.



Alternative text: a graphite hourglass runs, then empties beside one open-hand release card, which remains in a tray as the same unresolved cleanup obligation on later calls.

`MatchRomReadConversionStatus` is itself one allowlisted transaction. It performs one addressed copied read slot and reports one copied bit. A caller may schedule another such request only after the prior transaction and cleanup complete; the component hides no conversion poll loop.

Predict the result of calling `advance()` repeatedly after a timeout but before cleanup confirmation: number of successful phase receipts invented _____; cleanup quality _____.

8 Keep parasite power closed

Only **ExternallyPowered** requests are admitted at E0. **ParasitePower** returns **Unsupported** before state changes. There is no strong-pull-up phase or intent. Ordinary **Release** must never be reinterpreted as output-high drive or strong-pull power.

The data sheet explains that parasite operation uses the data line for power and imposes additional strong-pull requirements during operations such as temperature conversion. That is precisely why it is excluded here. A future parasite-power endpoint needs separate topology, current, rail, timing, exclusivity, rollback, stuck-on handling, and independent power-removal evidence.

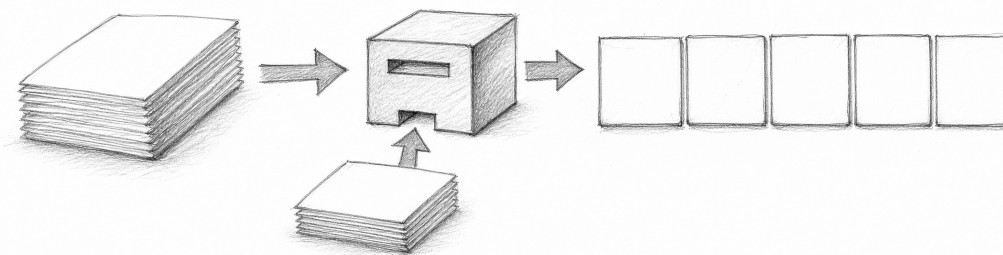
Request	E0 disposition	Reason
externally powered typed operation	eligible	copied protocol policy only
parasite-powered operation	Unsupported	no reviewed power capability
arbitrary captured transaction	invalid	outside closed operation surface
scratchpad EEPROM copy	absent	endurance and strong-pull work excluded

9 Replay the state machine

The compile-only Mega sketch supplies microsecond values, requests, receipts, and fixed caller-owned output buffers. It does not include a GPIO adapter and must not be uploaded as a bus driver.

```
OneWireStepIntent intent = {};
OneWireTransactionSnapshot snapshot = {};

OneWireTransactionPolicy transaction(config);
Status status = transaction.initialize(now, intent);
status = transaction.confirmCleanup(now, releaseReceipt);
status = transaction.begin(now, request, intent);
status = transaction.update(now, phaseReceipt, intent);
status = transaction.snapshot(snapshot);
```



Alternative text: a stack of graphite request cards and a separate receipt stack enter an abstract policy box, which fills five blank result cells in a row for the replay ledger.

The proof matrix covers every timing neighbor; presence absent, valid, early, late, and stuck; both copied read values; least-significant-first assembly; first, middle, and final slot failures; fieldwise changed duplicates; crossed correlation; single-drop restrictions; bounded search branches and collision; parasite rejection; timeout; cancellation; reset and shutdown cleanup; rollover and exhaustion; and two independent policies producing byte-stable results from the same trace.

10 Inspect gates before making a claim

Gate	E0 evidence	Still open
resources	zero hardware owners and exact compiler report	physical pin/timer/interrupt ownership
protocol	typed phases and copied timing boundaries	measured pulse, sample, rise, and recovery timing
cleanup	correlated Release intent and receipt	physical high-impedance/idle voltage
identity	byte-exact nonzero ROM transport	family, CRC, specimen authenticity
power	parasite requests reject atomically	any strong-pull topology or campaign

The ordinary Mega replay measures 9,032 bytes of flash and 763 bytes of static SRAM. The exact isolated no-LTO probe measures 10,182 bytes of flash, 763 bytes of static SRAM, and 215 bytes of conservative synchronous stack. The policy object is 232 bytes: an independently accepted miss above the 192-byte target and below the 256-byte hard limit. After the specified reserve and stack accounting, 7,086 bytes of Mega SRAM remain. Every hard and residual gate passes.

The ordinary sketch and exact isolated probe are separate measurement boundaries; their flash and static-SRAM values must not be added. These compiler measurements do not establish electrical timing, physical cleanup, or sensor behavior.

E1a remains open. Before any powered adapter, qualify an exact externally powered specimen and ordinary external pull-up. Record package and pinout, genuine/compatible evidence, supply and current, drive-low and release rails, rise time, reset/presence/slot waveforms, interrupt effects, timeout cleanup, stuck-low behavior, startup/reset/shutdown, and an independent physical line-release observation. Parasite power remains E1b-closed.

11 Completion record

Check	Result or evidence
Closed typed operation and nonzero-ROM transport boundary	
Every configured microsecond-window neighbor and rollover	
Intent/receipt correlation, duplicate, regression, and exhaustion	
Least-significant-first slots and attributable partial prefix	
Timeout, cancellation, reset, shutdown, and cleanup confirmation	
Parasite rejection and absence of hidden polling/retry	
Canonical replay, exact resource evidence, and open E1 gates	

Completion statement: I can distinguish copied protocol evidence from electrical execution, correlate every bounded phase, preserve partial prefix and cleanup debt, explain why release is not proof of an idle bus, and name the physical evidence still required.

Learner: _____ Reviewer: _____ Date: _____

Primary references

- Analog Devices, *DS18B20 Programmable Resolution 1-Wire Digital Thermometer*, Rev. 6, manufacturer data-sheet PDF.
- Analog Devices, *1-Wire Search Algorithm*, Application Note 187: manufacturer application-note page.
- Analog Devices, *1-Wire Extended Network Standard*, including its discussion of Application Note 126: manufacturer design-note page.

This print edition is untagged. Its text is searchable and copyable, and the HTML lesson provides the hardware-independent alternative. No PDF/UA, WCAG, or other PDF accessibility-conformance claim is made.